



Wire & cable: Optical devices, InP substrates

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Optical scale-up demand could be brought forward

We see major business opportunities for optical-related companies in both CPO and NPO

Lumentum Holdings talks about future demand for optical devices, InP substrates at briefing

Optical devices major Lumentum Holdings [LITE US] held a briefing for investors on 27 August that provided fresh evidence of the scope for growth in demand for the kinds of optical devices manufactured by Sumitomo Electric Industries [5802] (Buy) and Furukawa Electric [5801] (Buy), as well as the indium phosphide (InP) substrates that constitute a key input for these and that are primarily manufactured by Sumitomo Electric Industries and JX Advanced Metals [5016] (Buy).

Optical scale-up likely to be major positive for demand for both CPO and NPO optical devices

The share prices of Japan's three wire & cable companies have undergone numerous corrections since the start of 2026 on news about slow uptake of or delays to the deployment of co-packaged optics (CPO). However, US optical device and optical transceiver majors Lumentum Holdings, Coherent [COHR US], and Applied Optoelectronics [AAOI US] expressed different views when they released results in August. Coherent commented that it now expected demand to emerge earlier than its expectation three months earlier (*Figure 1*, see our 13 August 2026 report *Wire & cable: Earnings at non-Japanese companies (22)* and our 19 August 2026 report *Wire & cable: Earnings at non-Japanese companies (23)*).

Companies other than Nvidia more bullish about NPO than CPO

At investor briefings on 17 and 27 August, Lumentum Holdings commented that Nvidia had been securing substantial supplies of CPO components via long-term agreements, which was one of the reasons by other companies were being more bullish about near-packaged optics (NPO; *Figure 1*). We think recent comments about slow progress with CPO has been due to the proactive stance about NPO taken by companies other than Nvidia.

Scale-up represents completely fresh demand for optical-related companies in both CPO and NPO

At the investor briefing on 27 August, Lumentum Holdings said that whereas scale-out demand would involve replacing optical connections using existing optical transceivers with CPO, scale-up would require additional demand for optical connections overall, be they CPO or NPO. It then said that it saw greater business opportunities for NPO than for CPO in the short term, and that it saw potential for demand to emerge around one year earlier than its expectation prior to its results briefing. We think this is why Lumentum Holdings and other companies have said that CPO and NPO demand is being brought forward (*Figure 1*).

We also see substantial business opportunities in optical scale-up

While we think existing scale-out using CPO will be a positive for demand for optical devices and optical fiber, as it entails higher value-added and higher density optical connections than those using existing optical transceivers, we think scale-up will generate completely fresh business opportunities as it involves the replacement of copper connections with optical connections.

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Substantial new business opportunities in NPO too

Whereas CPO involves placing semiconductor chips and optical engines in the same package, NPO involves placing optical engines close to semiconductor chips but not in the same package. Because of this, NPO can often get away with slightly lower output from continuous wave laser diodes (CW-LDs), used as the external light source, than CPO, and NPO also features longer copper electrical wiring between the optical engine and the GPU and other semiconductor chips but shorter optical connections. However, it does involve replacing some of the previously 100% copper connections between chips in a rack with optical connections. We thus think that both CPO and NPO will result in substantial growth in demand for optical devices as electrical connections are replaced with optical connections. We also see rising demand for optical fiber as well as optical devices (*Figure 4*).

Lumentum Holdings says scale-across most underappreciated

Lumentum Holdings also said that scale-across was the most underappreciated application in the optical devices market. Whereas scale-out is primarily for connections between servers and between racks within a data center and scale-up is primarily for connections within racks, scale-across is about connections between data centers, ie data center interconnect (DCI). The company sees considerable latent demand for scale-across as part of the shift to building small power generation facilities and having geographically dispersed data centers to address the constraints on electric power supply to data centers.

We see growth in business opportunities for Sumitomo Electric Industries and other two wire & cable companies

The increased use of CPO and NPO in data centers should result in increased business opportunities for the three Japanese wire & cable companies Furukawa Electric, Sumitomo Electric Industries, and Fujikura [5803] (Buy). At its results briefing on 22 May, Sumitomo Electric Industries said that its optical devices for CPO were in the final stages of sample accreditation, which gave the impression that the company was slightly behind US rivals, but even so we still see major business opportunities from scale-up with CPO not only for optical devices but also for optical IC connection components. The company also commented that it was in discussions with Nvidia and multiple other customers, and we see potential for it to tap demand in NPO too (*Figure 1*). We also see business opportunities from scale-across for integrated tunable laser assemblies (iTLAs) used in long-distance optical transmission, where it has a high market share (see Figure 16 in our 3 August 2026 report *Sumitomo Electric Industries (5802 JP)*).

InP substrates: Lumentum Holdings constantly looking for suppliers

Lumentum Holdings also said at the 27 August conference that it was constantly looking for InP substrate suppliers as its long-term contracts with a Japanese company (we think this is probably Sumitomo Electric Industries) and AXT [AXTI US] may not provide enough supply. We think this could result in fresh business opportunities for Japanese companies Sumitomo Electric Industries and JX Advanced Metals (*Figure 6*, *Figure 7*).

Sumitomo Chemical starting mass production of InP epiwafers, we think JX Advanced Metals is substrate supplier

On 31 August, Sumitomo Chemical [4005] (Neutral) announced the start of mass production and sale of InP epiwafers, and said it was aiming for sales of around ¥10bn by the mid-2030s. While we see limited impact on overall core operating profits at the company (¥208.4bn in 26/3), we think this demonstrates strong demand for stable procurement of InP epiwafers across the industry. InP epiwafers are manufactured by stacking layers (epitaxial layers) onto an InP substrate. Sumitomo Chemical is mass-producing InP epiwafers at its Ibaraki Plant (in the city of Hitachi, Ibaraki Prefecture), while JX Advanced Metals is planning to increase output of InP substrates at its Isohara plant (in Kitaibaraki, Ibaraki Prefecture) and at its Hitachinaka works (in Hitachinaka, Ibaraki Prefecture). Accordingly, we estimate that JX Advanced Metals may be supplying InP substrates to Sumitomo Chemical, although Sumitomo Chemical has not said anything about this.

Fig. 1: Companies' sales and production outlooks for CPO and NPO products

Code	Company	Business overview	Timing (yy/m/d)	Events	Comments on CPO and NPO
GLW US	Corning	Optical fiber, etc	26/5/6	Investor briefing	The scale-out market could start as early as 2027 and become full-fledged in 2028, and the market could grow to around \$10.0bn in 2030 on sharp growth in demand, including for scale-up switches.
5802 JP	Sumitomo Electric Industries	Optical devices, optical connectors, optical fiber, etc	26/5/22	Results briefing	High-output continuous-wave laser diodes (CW-LDs) are currently undergoing final sample certification. Discussions are in progress with Nvidia and several other customers. Pilot shipments are scheduled to start at end-27/3 and ramp up in 28/3. We expect sales related to optical fiber and connection components to be booked on a small scale from 28/3.
AAOI US	Applied Optoelectronics	Optical transceivers, optical devices	26/8/7	Results briefing	It aims to increase output of External Laser Small Form-Factor Pluggable (ELSFP) modules (which include high-power lasers) between 2026 H2 and 2027 and increase output to 400,000 units a month in 2028. Ramping up mass production of high-power lasers takes 21–24 months or more from the placement of equipment orders. Demand is currently substantially higher than supply capacity.
LITE US	Lumentum Holdings	Optical devices, optical transceivers	26/8/11	Results briefing	The company expects demand for its ultra high-power laser chips for CPO applications to ramp up in 2027 H2. The company has received ELSFP orders for delivery through 2027. It plans to start major shipments for CPO applications in 2027 H2. Shipments to major customers for NPO applications could begin one quarter earlier than for CPO applications.
COHR US	Coherent	Optical transceivers, optical devices	26/8/12	Results briefing	The company expects ultra-high power-continuous-wave laser diodes (CW-LD) for CPO solutions for scale-out applications to ramp up in Oct–Dec 2026 and lasers for CPO for scale-up applications to ramp up in 2027 H2. The company noted that it is seeing no sign whatsoever of CPO demand being pushed back, and, on the contrary, it says customer requests are being pulled forward.
LITE US	Lumentum Holdings	Optical devices, optical transceivers	26/8/17	Investor briefing	The company expects the scale-up market to get off the ground in 2028, with CPO accounting for half, mainly for Nvidia. NPO will account for the remainder, for other others. The number of lasers per GPU in scale-up is 5–10x that in scale-out.
LITE US	Lumentum Holdings	Optical transceivers, optical devices	26/8/27	Investor briefing	Before its latest results briefing, the company had thought that NPO demand would begin in late 2028 or the start of 2029, but it now thinks NPO customers will deploy the solutions sooner because of competition with Nvidia's CPO, and expects demand to begin from late 2027 through the beginning of 2028. It sees greater business opportunities in NPO than in CPO in the near term.

Note: NPO stands for Near-Packaged Optics.

Source: Data disclosed by each company, Nomura

Fig. 2: Major long-term contracts for optical products (2026) (1)

According to official company announcements

Code	Company	Announcement date (yyyy/m/d)	Customers (partners)	Period (years)	Contract value	Products covered
Official announcement						
GLW US	Corning	2026/1/27	Meta Platforms	Multiple years	Up to \$6bn	Optical fiber, optical cable, optical connectivity solutions, etc
LITE US	Lumentum Holdings	2026/3/2	NVIDIA	Multiple years	Several billion dollars	Optical laser products
COHR US	Coherent	2026/3/2	NVIDIA	Multiple years	Several billion dollars	Optical laser products, optical network products
GLW US	Corning	2026/4/28	Two HSDCs	Multiple years	NA	Optical fiber, optical cable, optical connectivity solutions, etc
GLW US	Corning	2026/5/6	NVIDIA	Multiple years	NA	Optical connectivity solutions, optical fiber
GLW US	Corning	2026/6/8	Amazon	Multiple years	Several billion dollars	Optical fiber, optical cable, optical connectivity solutions
PRY IM	Prysmian	2026/7/20	Molex	10 years	Up to €5.5bn	Supply of optical cable for use in data centers
SOTL IN	Sterlite Technologies	2026/7/24	One HSDC	27/3–29/3	\$1.1bn	Optical connection products
5801 JP	Furukawa Electric	2026/8/4	NA	NA	NA	Optical fiber, optical cable
GLW US	Corning	2026/8/20	Zayo	4.5 years (through 2030)	NA	Optical fiber, optical cable

Note: (1) Corning announced on 28 April 2026 that it had entered agreements with two hyperscalers, and that the scale and duration of these agreements were similar to the agreement with Meta Platforms announced in January. (2) HSDC = hyperscaler.

Source: Company data, Nomura estimates

Fig. 3: Major long-term contracts for optical products (2026) (2)

Briefing comments, long-term raw material procurement contracts

Code	Company	Announcement date (yyyy/m/d)	Customers (partners)	Period (years)	Contract value	Products covered
Briefing comments						
5802 JP	Sumitomo Electric Industries	2026/2/3	NA	3–7 years	NA	Products in infocommunications segment
PRY IM	Prysmian	2026/4/30	HSDC	3–7 years	NA	Optical fiber, optical cable
COHR US	Coherent	2026/5/6	Multiple HSDCs	Multiple years	NA	CPO-related products
5802 JP	Sumitomo Electric Industries	2026/7/10 (Interview with company president)	One HSDC	3 years	NA	Optical fiber cable
COHR US	Coherent	2026/8/12	Multiple customers	Multiple years (many through end-2030)	NA	Optical transceivers, CPO lasers (estimated)
Input procurement contract comments						
LITE US	Lumentum Holdings	2026/3/17	Japanese suppliers	2007	NA	Indium phosphide (InP) substrates
5803 JP	Fujikura	2026/5/14	NA	Multiple years	NA	Optical fiber, optical fiber preform
COHR US	Coherent	2026/7/2	AXT	3 years	NA	Six-inch InP substrates
LITE US	Lumentum Holdings	2026/7/29	AXT	Through end-2031 (5 years and 4 months)	NA	InP substrates

Note: (1) Prysmian said at its 30 April 2026 briefing that it planned to sign long-term (3–7 year) optical cable contracts with hyperscalers over the next few weeks. (2) HSDC = hyperscaler.

Source: Company data, Nikkei, and Nomura

Fig. 4: Opportunities for growth in market for optical fiber and wiring materials at data centers

Based on Corning briefing materials released on 6 May 2026

Themes	Required amount of optical fiber and wiring materials	Details
GPU cluster expansion	1.5x	If the number of GPUs in a single cluster exceeds 130,000, three layers are needed for the cluster, as two would be insufficient. This would result in a roughly 50% increase in demand. The number of three-layer clusters has likely gradually increased since 2025.
Data transmission capacity (bandwidth) expansion	2–4x	Nvidia's specs use single-wave 100G EMLs in both 400G and 800G products, so optical fiber volume would double in the transition from 400G to 800G. As single-wave 200G EMLs are used in 1.6T products, there is no change in fiber count in the transition from 800G to 1.6T. With 3.2T products, the fiber count would double if single-wave 200G EMLs are used, but it wouldn't change if single-wave 400G EMLs are used. The shift from 400G to 3.2T would involve a change in the fiber count of anywhere from 2x to 4x. If optical fiber volumes were stay the same, this would make optical devices more sophisticated.
Optical scale up	10x	At present, all wiring in racks is copper wire, but if all wiring in racks were to be optical, the number of fiber needed per GPU would be 10x the current level.
Passive photonics	New demand	New CPO/NPO business opportunities. Fibers and other components connecting external light sources and other elements to optical engine and other components on silicon photonics chips. New business opportunities seen as amounting to \$10bn by 2030.

Source: Nomura, based on Corning materials

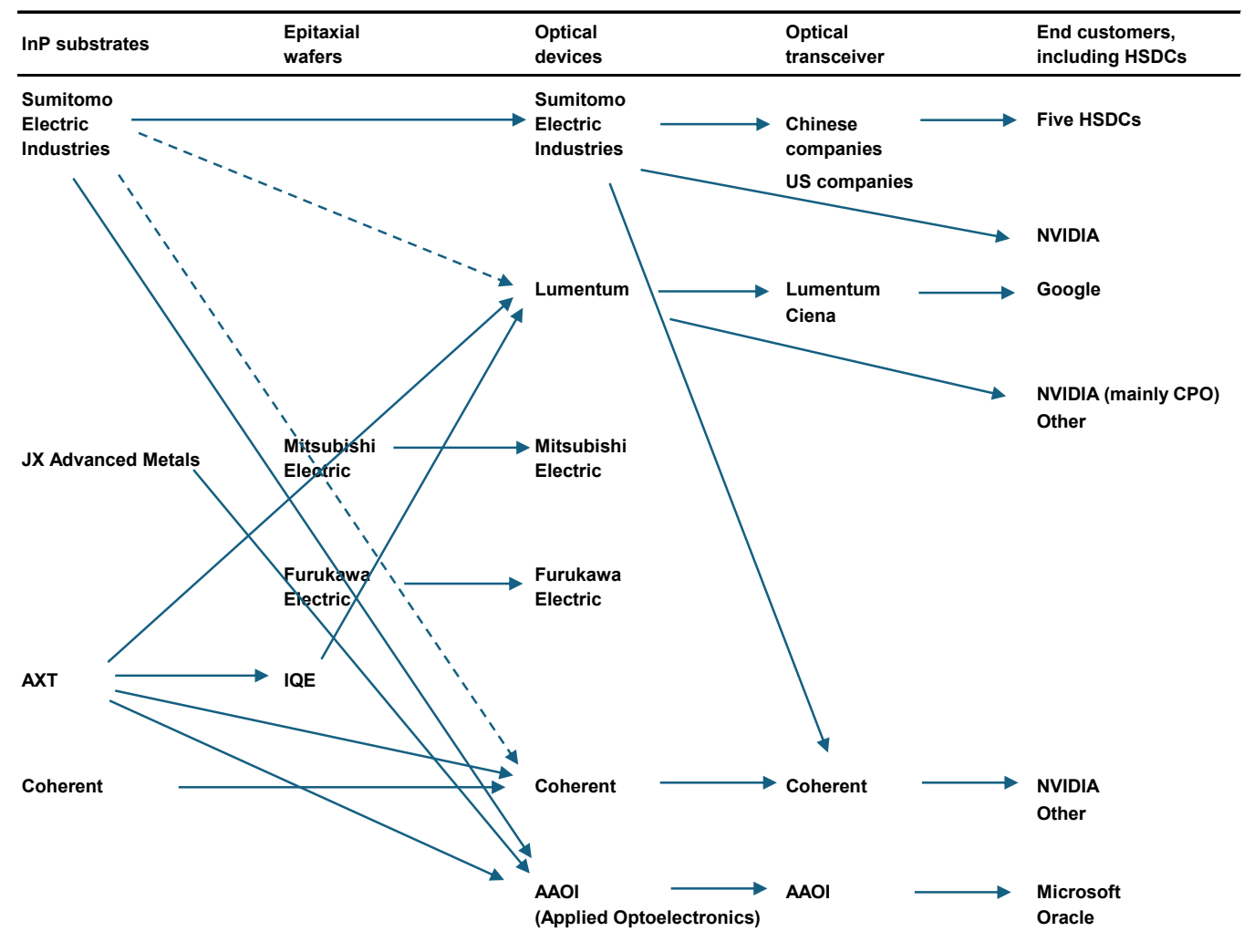
Fig. 5: Sumitomo Electric Industries' products that can be used in co-packaged optics

Product category	Products	Details	Competitors
Optical IC connection component	Two-dimensional optical fiber array with 90-degree bend (2D-FBGE (FlexBeamGuidE))	Optical connection component used in CPO switches. Fabricated by assembling a 90-degree bent optical fiber with a small radius into a two-dimensional glass-hole array. 2D-FBGE has a low profile of 5.5mm, high density of 24 fibers/mm, low insertion loss of less than 0.5dB, and high polarization-maintaining characteristics of 20dB or more.	Corning (US Conec), Senko Advance
CW-LD	400mW-class 1.3μm high-power semiconductor laser	Whereas EML output is a only few mW , CW-LD output is at least 100mW. External laser sources (ELS), a key component of CPO, require high output of several hundred mW and power conversion efficiency of at least 20%. This product has a high output of over 400mW and a power conversion efficiency of 25%.	Lumentum Holdings, Coherent, Furukawa Electric

Note: (1) Corning has a 50% stake in US Conec and Fujikura has a 30% stake. (2) Senko Advance is an unlisted Japanese company. We assume that Furukawa Electric supplies mechanical transfer (MT) ferrules.

Source: Nomura based on company materials (Sumitomo Electric Technical Review for January 2023 and July 2024)

Fig. 6: Supply chain for optical devices (1)



Note: (1) Nomura estimates, based on briefings and press releases. See figures below for details. (2) HSDC = hyperscaler. (3) According to Lumentum Holdings, it has a long-term agreement for InP substrates with one Japanese company. This is either Sumitomo Electric Industries or JX Advanced Metals. At its 3 February 2026 results information session, Sumitomo Electric Industries commented that it had contracts ranging from three years to a maximum of seven years in its infocommunications segment, though it did not disclose the specific product groups. Lumentum Holdings also said that it had signed a seven-year InP substrate procurement agreement with one supplier in Japan. A seven-year long-term agreement is not very common (see our 31 March 2026 report Sumitomo Electric Industries [5802]). Given this, and the fact that a 10 April 2026 Bloomberg interview with Lumentum Holdings' Michael Hurlston cited Fujikura and Sumitomo Electric Industries as Japanese companies with strengths in InP substrates, optical fiber materials, and silicon photonics, we surmise that Sumitomo Electric Industries is the likely party to the seven-year agreement. (4) Solid lines show actual links based on company announcements and briefing comments, dotted lines show Nomura estimates for highly likely connections.

Source: Nomura, based on data disclosed by each company and Bloomberg data

Fig. 7: Supply chain for optical devices (2)

Code	Company	(yy/m/d)		Company comments, etc
5802 JP	Sumitomo Electric Industries	24/11/13	Results briefing	Company mentioned relationships with North American transceiver companies including Coherent
		25/3/18	Nvidia press release	Selected as a supplier of silicon photonics, including for Nvidia's CPOs
		26/7/10	Comment by company president (Nikkei)	Noted that its products are used in telecommunications equipment supplied to five North American HSDCs (given the company's high CW-LD market share and other factors, we estimate that it supplies Chinese transceiver companies.)
LITE US	Lumentum HD	25/12/8	IQE release	Long-term multi-year epitaxial wafer contract with IQE
		26/3/2	Release	Long-term multi-year optical laser component contract with Nvidia worth several billion dollars
		26/4/10	CEO comment (Bloomberg)	Regarding InP substrates, the company mentioned an extremely important seven-year, long-term agreement with one supplier in Japan
		26/5/5	Results briefing	Google is indisputably the largest customer. 20% of CW-LDs used in the company's optical transceivers are manufactured in-house.
		26/7/30	AXT release	Long-term InP substrate agreement with AXT through end-2031
		26/8/11	Results briefing	On the topic of InP substrates: We're probably pretty comfortable just given our lead in arrangement with the Japanese supplier and now of the announced deal with AXTI
		26/8/17	10-K	In 26/6, Google and Ciena accounted for 26.6% and 16.0% of sales
		26/8/17	Investor briefing	Much of the In substrates made by our supplier in Japan are for us as we have seven-year supply agreement with them
		26/8/17	Investor briefing	1.6tn optical transceivers are supplied to two major HSDC customers
COHR US	Coherent	26/2/4	Results briefing	The company said it has secured multiple suppliers for 6-inch InP substrates (Sumitomo Electric and AXT are the only suppliers for 6-inch substrates, so we assume it does business with Sumitomo Electric Industries)
		26/3/2	Release	Multi-year, multi-billion-dollar contract with Nvidia to supply optical laser components and optical network equipment
		26/7/2	AXT release	Three-year, long-term agreement with AXT for procurement of 6-inch InP substrates
		26/8/12	Results briefing	Said it does not sell its in-house produced optical devices externally.
AAOI US	Applied Optoelectronics	26/2/26	10-K	The largest component of 25/12 sales was Microsoft (28.8%). Microsoft accounted for 46.6% of sales in 24/12, followed by Oracle (12.4%).
		26/8/7	Results briefing	On the topic of InP substrates, the company said it has locked in two European suppliers, two Japanese suppliers, and three Chinese suppliers
		26/8/18	Investor briefing	The main customers for 800G transceivers are two HSDC companies. Small-scale sales and sales likely to start soon for several customers.
		26/8/18	Investor briefing	Five companies in discussions about CPOs, some of which are in discussions about NPOs

Note: (1) Company comments are indicated or provided in quotes. Figures in parentheses are our estimates. (2) HSDC = hyperscaler. (3) According to Lumentum Holdings, it has a long-term agreement for InP substrates with one Japanese company. This is either Sumitomo Electric Industries or JX Advanced Metals. At its 3 February 2026 results information session, Sumitomo Electric Industries commented that it had contracts ranging from three years to a maximum of seven years in its infocommunications segment, though it did not disclose the specific product groups. Lumentum Holdings also said that it had signed a seven-year InP substrate procurement agreement with one supplier in Japan. A seven-year long-term agreement is not very common (see our 31 March 2026 report on Sumitomo Electric Industries [5802J]). Given this, and the fact that a 10 April 2026 Bloomberg interview with Lumentum Holdings' Michael Hurlston cited Fujikura and Sumitomo Electric Industries as Japanese companies with strengths in InP substrates, optical fiber materials, and silicon photonics, we surmise that Sumitomo Electric Industries is the likely party to the seven-year agreement.

Source: Nomura, based on company, Nikkei, and Bloomberg data

Fig. 8: Wire & cable: Events including results announcements by Japanese and overseas companies

Date	Time	Code	Company	Event	Description of business	Key points	Related stocks
26/9/1	17:00	CRDO US	Credo Technology	Results announcement	Active electrical cable and other connectivity solutions	Copper wire and optical connectors in servers and racks	Fujikura, Furukawa Electric, Sumitomo Electric Industries
26/9/2	After close of trading	AVGO US	Broadcom	Results announcement	Semiconductor	AI semiconductor demand	Fujikura, Furukawa Electric, Sumitomo Electric Industries
26/9/3	8:30	CIEN US	Ciena	Results announcement	Optical communications equipment	Demand for coherent optical transceivers etc	Sumitomo Electric Industries, Furukawa Electric
26/9/9	08:10	GLW US	Corning	Citi's TMT Conf	Optical fiber-related (US)	Demand for optical fiber and optical wiring components in North America	Fujikura, Furukawa Electric, Sumitomo Electric Industries
26/9/9	14:35	LITE US	Lumentum Holdings	Citi's TMT Conf	Optical devices, optical transceivers	Optical device demand	Sumitomo Electric Industries, Furukawa Electric
26/9/10	NA	AXTI US	AXT	B. Riley Securities Consumer & TMT Conf	Compound semiconductors	Demand for indium phosphide substrates	Sumitomo Electric Industries
26/9/10	NA	NVDA US	NVIDIA	Goldman Sachs Communacopia and Tech Conf	Semiconductor	AI semiconductor demand	Sumitomo Electric Industries, Fujikura, Furukawa Electric
26/9/10	13:50	CRDO US	Credo Technology	Goldman Sachs Communacopia and Tech Conf	Active electrical cable and other connectivity solutions	Copper wire and optical connectors in servers and racks	Fujikura, Furukawa Electric, Sumitomo Electric Industries
9/20-24	NA	NA	EOCO (European Conference and Exhibition on Optical Comunucations) 2026 , Malaga, Spain			Global market trends for optical products	Fujikura, Furukawa Electric, Sumitomo Electric Industries
9/21-22	NA	AXTI US	AXT	Morgan Stanley ASIA Best Corporate Day	Compound semiconductors	Demand for indium phosphide substrates	Sumitomo Electric Industries
26/9/29	NA	NKT DC	NKT	Investor Day	Electric power cable (Europe)	Electric power cable demand in Europe	Sumitomo Electric Industries
26/10/6	Morning	MRVL US	Marvell Technology	Investor Day	Semiconductor	AI semiconductor demand	Fujikura, Furukawa Electric, Sumitomo Electric Industries

Note: Earnings results announcement times are local times. EST for US companies.

Source: Nomura, based on data disclosed by each company